TEXORA · LECTURE SHEET

Quality Management in Apparel Industry

From inspection to management system: the evolution inspection → QC → QA → TQM, ISO 9001 in a garment factory, the cost of quality, statistical process control with a worked p-chart, and the audit culture that buyers actually buy.

AM-4201 · ADVANCED APPAREL MANUFACTURING-II · TOPIC 8 · LEVEL 4 TERM 2

1. The evolution: four stages of an idea

STAGE	CORE IDEA	IN APPAREL TERMS
Inspection	Sort good from bad at the end.	End-of-line checking and the final audit — necessary, but it only finds faults the process already made (TE-3107).
Quality control (QC)	Measure the process and correct deviations.	In-line checks, DHU charts, AQL sampling, SPC (§4): control moves upstream to where defects are born.
Quality assurance (QA)	Design the system so faults cannot easily occur.	Documented procedures, sealed standards, calibrated equipment, trained and certified operators, approved-supplier systems.
TQM	Quality is everyone's job, improved continuously, aimed at the customer.	Whole-factory culture: line operators own quality, PDCA cycles run weekly, buyer requirements drive internal standards.

The stages accumulate rather than replace: a TQM factory still inspects — but inspection is now the audit of a system designed to prevent, not the net that catches a careless process.

2. ISO 9001 in a garment factory

ISO 9001 certifies a **quality management system**, not a garment: auditors verify that the factory says what it does (documented procedures), does what it says (records), and improves (corrective actions, management review). Its process approach and PDCA cycle (Plan–Do–Check–Act) organise the apparel-specific elements this course has built one by one: document control (tech packs, operation bulletins — Topic 4), purchasing control (supplier records — Topic 6), production control (SMV standards — Topic 3), monitoring and measurement (inspection systems — TE-3107, AM-4102 Study 3), control of nonconforming product (defective lots held, sorted, dispositioned) and corrective action (root cause, not blame). Buyers layer their own technical audits on top: needle control, measurement systems, wash standards, lab testing — a factory's quality management is judged by the file it can show and the floor that matches it.

3. The cost of quality

COST FAMILY	APPAREL EXAMPLES	BEHAVIOUR
Prevention	Training, method study, pattern/marker checking, supplier approval, maintenance.	Cheapest taka in the system; spending here shrinks both failure families.
Appraisal	Inspectors, audits, lab tests, needle detectors, measurement.	Necessary; grows when prevention is weak.
Internal failure	Rework, repair, recuts, seconds, line stoppage, re-audit.	Found before shipment — expensive in minutes and cloth.
External failure	Claims, chargebacks, air freight to recover dates, rejected shipments, lost orders.	The catastrophe family: one failed buyer audit can cost more than a year of prevention.

The economics argue the TQM case without philosophy: quality spending moved from failure to prevention lowers total quality cost *and* raises quality — the only free lunch in management. Factories track the ratio (cost of quality as % of sales) and its family mix as a board-level KPI (Topic 1).

4. SPC on the sewing floor: a worked p-chart

The p-chart plots the fraction defective in periodic samples and distinguishes **common-cause** variation (the process's normal noise, inside control limits) from **special causes** (a point outside limits —

something changed; find it). Reacting to noise as if it were signal (and vice versa) is the twin error SPC exists to prevent.

WORKED P-CHART — VERIFY EVERY NUMBER

Daily final-check samples of n = 200 garments give defectives: Day 1: 12 · Day 2: 9 · Day 3: 14 · Day 4: 11 · Day 5: 30. Fractions: 0.060, 0.045, 0.070, 0.055, 0.150. Centre line $\bar{p}$ = (12 + 9 + 14 + 11 + 30) ÷ 1,000 = **0.076**. Control limits: $\bar{p} \pm 3\sqrt{[\bar{p}(1 - \bar{p}) \div n]}$ = 0.076 ± 3√(0.076 × 0.924 ÷ 200) = 0.076 ± 3 × 0.0187 = 0.076 ± 0.056 → **UCL = 0.132, LCL = 0.020**. Days 1–4 sit inside the limits — common-cause variation; do not chase individual days. Day 5 (0.150 > 0.132) is **out of control**: a special cause entered (investigation found a new operator group and a blunt needle batch that morning — exactly the class of cause a chart is meant to trigger a hunt for). Action: contain Day 5 production, fix causes, and only then recalculate limits from clean data.

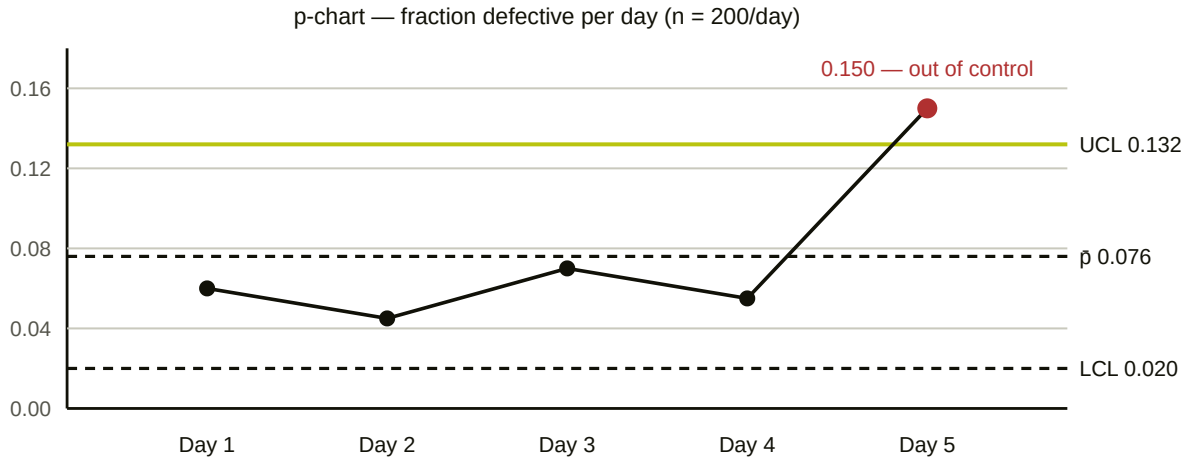


Figure 1 — The worked p-chart, plotted to scale (0 at y = 210; 0.01 of fraction = 10 px; centre line 0.076, UCL 0.132, LCL 0.020). One point in five carries the message.

5. Continuous improvement as routine

TQM's engine is the PDCA cycle run at every level: a line team charts DHU, picks the top defect (Pareto), finds the root cause (the diagnostic method of TE-3107 Topic 4), changes the method, and re-charts — the factorial method of Topic 7 is its heavy artillery for process questions. Quality circles, 5S housekeeping and visual management are the cultural carriers; the audit calendar (internal audits, buyer audits, certification surveillance) is the external discipline. The apparel-specific truth closing this course: in an industry where the product changes weekly, quality cannot be a department's output — it is the operating system of the whole factory, or it is a certificate on a wall.

Glossary

TERM	MEANING
QMS	Quality management system: the documented policies, processes and records by which an organisation directs quality (ISO 9001).
PDCA	Plan–Do–Check–Act: the improvement cycle at the heart of ISO 9001 and TQM.
p-chart	Control chart for fraction defective in samples; limits at $\bar{p} \pm 3\sqrt{[\bar{p}(1-\bar{p})/n]}$.
Common / special cause	Variation inherent to the process / variation from an identifiable disturbance; charts exist to tell them apart.
Cost of quality	Prevention + appraisal + internal failure + external failure costs, tracked as a management ratio.

Key points and summary

- Inspection finds, QC controls, QA prevents by system, TQM makes quality everyone's continuous job.
- ISO 9001 certifies the management system; buyers audit the floor and the file together.
- Moving spending from failure to prevention lowers total quality cost while raising quality.
- The p-chart separates noise from signal: worked limits UCL 0.132 / LCL 0.020 around $\bar{p}$ 0.076; one point outside is an investigation order.

Practice

Answer in your own words; brief model answers follow.

1. Samples of n = 250 give defectives 15, 20, 18 over three days. Compute $\bar{p}$ and the 3-sigma limits.
2. Classify by quality-cost family: (a) operator training course; (b) final auditor's salary; (c) re-sewing 300 garments after an audit fail; (d) a buyer's chargeback for late shipment after quality rejection.
3. Why should limits not be recalculated immediately after an out-of-control point is found?
4. Give one apparel example each of a prevention activity replacing an appraisal activity, and of appraisal replacing failure.
5. Why does ISO 9001 certification not guarantee that a factory's garments are good?

Answer key

1. Total defectives = 53, total inspected = 750 → $\bar{p}$ = 53 ÷ 750 = 0.0707. $\sigma = \sqrt{(0.0707 \times 0.9293 \div 250)} = \sqrt{0.000263} = 0.0162$. Limits = 0.0707 ± 3 × 0.0162 = 0.0707 ± 0.0486 → UCL = 0.119, LCL = 0.022.

2. (a) Prevention. (b) Appraisal. (c) Internal failure. (d) External failure.
3. The out-of-control data describe a disturbed process; folding them into the limits widens the limits around the disturbance and normalises it. First remove the special cause and confirm stability, then recalculate from clean data.
4. Prevention replacing appraisal: operator certification and method standardisation at a problem operation reduce the need for 100% checking there to patrol checks. Appraisal replacing failure: in-line checking at the operation catches defects as singles instead of the final audit failing a whole lot into sorting and rework (internal failure) or the buyer rejecting it (external failure).
5. It certifies that a management system exists, is documented and is followed — a factory can consistently follow a mediocre system. Garment quality depends on the standards the system enforces (buyer specs, AQL levels, test regimes) and the competence inside it; hence buyers run their own technical audits beyond the certificate.

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